10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 2 — Human Respiratory System

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. Breathing means:

(a) Breakdown of C–H bonds to produce energy

(b) Movements that take air in and out of body

(c) Getting oxygen from the air and removing carbon dioxide

(d) Transport of oxygen through blood to different parts of the body

Answer: (b) — Breathing is a physical process in which air is moved into the lungs and expelled out of them. The breakdown of C–H bonds is cellular respiration (a chemical process), and the transport of oxygen through blood is carried out by the circulatory system.

2. Which path does air take immediately after passing through the larynx?

(a) Pharynx

(b) Bronchi

(c) Trachea

(d) Alveoli

Answer: (c) — Air enters through the nostrils into the nasal cavity, then goes to the pharynx, larynx, trachea (windpipe), bronchi, bronchioles and finally the alveoli. So the trachea comes immediately after the larynx.

3. Why are alveoli surrounded by capillaries?

(a) To absorb water

(b) To exchange gases

(c) To produce mucus

(d) To trap dust particles

Answer: (b) — Oxygen has to move from the air in the alveoli into the blood, and carbon dioxide has to move from the blood into the air. The capillaries around each alveolus make this exchange of gases possible.

4. Which feature of alveoli helps in rapid gas exchange?

(a) Thick walls

(b) Large diameter

(c) Moist surface

(d) Strong muscles

Answer: (c) — The alveoli are thin-walled pouches whose surface is moist, so oxygen dissolves in this moisture and then diffuses into the blood quickly. Their walls are only one cell thick, which also makes gaseous exchange rapid.

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5. Which action increases lung volume during breathing?

(a) Relaxing the diaphragm (b) Contracting intercostal muscles
(c) Closing the trachea (d) Stiffening alveolar walls

Answer: (b) — During inhalation the intercostal muscles contract and move the ribcage upward and outward, and the diaphragm also contracts and moves down. These two contractions increase the volume of the chest, so air rushes into the lungs.

6. Inhaled air contains 21 % oxygen. What is its percentage in exhaled air?

(a) 0 % (b) 16 %
(c) 21 % (d) 42 %

Answer: (b) — Some oxygen of the inhaled air is absorbed by the blood in the lungs, so the exhaled air contains about 16 % oxygen. Its carbon dioxide increases from about 0.04 % to about 4 %, while nitrogen remains 79 % in both.

7. Which factor controls the breathing rate?

(a) Carbon dioxide in blood (b) Oxygen in blood
(c) Carbon dioxide in muscles (d) Oxygen in muscles

Answer: (a) — The respiratory centre in the brain detects a greater concentration of carbon dioxide in the blood. During exercise more carbon dioxide is produced, so the centre sends messages to the muscles of the ribs and diaphragm at greater speed and the breathing rate increases from 16–20 to 30–40 times per minute.

8. The walls of alveoli are broken in:

(a) Pneumonia (b) Bronchitis
(c) Emphysema (d) Asthma

Answer: (c) — In emphysema the inner walls of the alveoli become weak and then rupture, so one large air space is formed instead of many small ones. In this way the surface area for gaseous exchange decreases and the patient cannot breathe properly.

9. Smoking leads to lung cancer by:

(a) Increasing oxygen uptake (b) Promoting uncontrolled cell division
(c) Slowing down healing (d) Enlarging the trachea

Answer: (b) — Cancer means an uncontrolled increase in the number of cells in a part of the body. Tobacco smoke contains more than 69 known carcinogens, such as tar and benzene, which promote uncontrolled cell division, so masses of non-functional cells are formed in the lungs.

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10. Which chemical in cigarette smoke directly leads to addiction?

(a) Carbon monoxide (b) Tar

(c) Nicotine (d) Formaldehyde

Answer: (c) — Nicotine is the major component of cigarettes and the chemical which makes a person addicted to smoking. It also damages many tissues including the brain, and it hardens the walls of the arteries.

B. Write short answers.

1. Differentiate between gaseous exchange and breathing.

Gaseous exchange	Breathing
It is the movement of oxygen from the inhaled air into the blood and of carbon dioxide from the blood into the air.	It is the movement of air into and out of the lungs.
It takes place in the alveoli of the lungs, through their thin, moist walls.	It takes place by the movements of the ribs and the diaphragm.
It is a diffusion process, so it is a physical process that happens at the surface of the alveoli.	It is a physical process of inhaling and exhaling air.
It supplies oxygen to the blood and removes carbon dioxide from it.	It brings fresh air into the lungs and takes stale air out, so that gaseous exchange can take place.

2. Differentiate between breathing and cellular respiration.

Breathing	Cellular respiration
It is a physical process in which air is taken in and given out.	It is a chemical process in which cells oxidize food (glucose) and get energy.
It takes place in the lungs and the air passageway.	It takes place in the cells of the body.
No energy is produced in this process; energy is only used in the movement of muscles.	Energy (ATP) is produced in this process.
It only helps in gaseous exchange, which in turn helps in respiration.	It uses the oxygen supplied by gaseous exchange and produces carbon dioxide and water.

3. What is the function of cartilages present in the walls of trachea and bronchi?

In the walls of the trachea, C-shaped cartilages are present, while the walls of the bronchi have plate-like cartilages. These cartilages are strong but flexible, so they keep the trachea and the bronchi open and prevent them from collapsing even when there is no air in them. In this way the air passage remains open during both inhalation and exhalation. Cartilage is not present in the smallest bronchioles, which are kept open by the pressure of the air inside them.

4. What is the average breathing rate in human at rest and during exercise?

In normal conditions, that is at rest, a human breathes about 16–20 times per minute. During exercise or hard work the muscles carry out cellular respiration at a greater rate, so more carbon dioxide is produced. The respiratory centre in the brain detects this and sends messages faster to the muscles of the ribs and diaphragm, so the breathing rate increases up to 30–40 times per minute.

5. How does the structure of alveoli maximize gas exchange?

The alveoli are thin-walled pouches, and each alveolus is surrounded by a network of blood capillaries. Their walls are only one cell thick and their inner surface is moist, so oxygen dissolves in the moisture and diffuses quickly into the blood, while carbon dioxide diffuses out of the blood into the air. There are millions of alveoli in the lungs, which together provide a very large surface area for gaseous exchange. This thin, moist and very large surface makes the exchange of gases very rapid.

6. List the symptoms, causes, and treatments of bronchitis.

Bronchitis means inflammation of the bronchi or bronchioles, in which the inner space of the bronchioles decreases. It may be acute (1–2 weeks) or chronic (3 months to 2 years).

Feature	Detail
Causes	Infection by viruses or bacteria, and chemical irritants such as tobacco smoke.
Symptoms	Coughing, wheezing, fever, chills and shortness of breath.
Treatment	Medicines (cough suppressants) to ease breathing, antibiotics if the infection is bacterial, antiviral medicines if it is viral, and avoiding irritants like smoke and pollution.

7. Mention the symptoms and treatments of emphysema.

Symptoms: shortness of breath, chronic cough (with or without mucus), wheezing and chest tightness. In this disease the inner walls of the alveoli become weak and rupture, so one large air space is formed instead of many small ones and the surface for gaseous exchange decreases.

Treatment: medicines are given to relieve cough and breathing problems, and the patient is asked to avoid tobacco smoke, chemical fumes and other air pollutants.

Emphysema cannot be cured completely, because the broken walls of the alveoli do not grow again.

8. What are carcinogens? Name any two carcinogens present in cigarettes.

Carcinogens are those substances which cause cancer, that is, they promote uncontrolled cell division in the body. Tobacco smoke contains more than 4,000 different chemicals, out of which more than 69 are known carcinogens. Two of them present in cigarettes are **tar** and **benzene**; formaldehyde is also a carcinogen present in cigarette smoke.

9. Complete the following table: (Feature, Inspired Air, Expired Air — Oxygen, Carbon dioxide, Nitrogen)

Feature	Inspired (inhaled) air	Expired (exhaled) air
Oxygen	21 %	16 %
Carbon dioxide	0.04 %	4 %
Nitrogen	79 %	79 %

C. Write answers in detail.

1. Explain the process of inhalation and exhalation step by step.

Breathing is a physical process in which air is taken into the lungs and expelled out of the lungs. It has two steps: inhalation and exhalation.

1. Inhalation (inspiration): inhalation means taking air into the lungs. The intercostal muscles present between the ribs contract, which moves the ribcage upward and outward. At the same time the diaphragm, a thick muscular sheet below the lungs, also contracts and moves down. These two contractions increase the volume of the chest cavity. As the volume increases, the pressure inside the chest is reduced, so air from outside rushes into the lungs through the air passageway.

2. Exhalation (expiration): exhalation means moving the air out of the lungs. The intercostal muscles now relax, so the ribcage moves inward and downward. At the same time the diaphragm relaxes and moves up. These movements decrease the volume of the chest cavity, so the pressure inside the chest increases and the air is forced out of the lungs through the same passageway.

In this way the alternate contraction and relaxation of the intercostal muscles and the diaphragm carry on breathing, and about half a litre of air is exchanged in each breath. The air we breathe out contains water vapour, which is the moisture we see on a glass when we breathe on it.

2. Write a short note on the major parts of the human air passageway.

The connected tubes through which air passes to the lungs are collectively called the air passageway. Its main parts are:

1. Nostrils and nasal cavity: air enters through the nostrils into the nasal cavity, which is made of two narrow cavities between the nostrils and the pharynx. Fine hairs filter the incoming air by trapping large dust particles, mucus traps the small dust particles, and the mucus also warms the air so that its temperature becomes

nearly equal to the body temperature.

2. Pharynx: the nasal cavity opens into the pharynx, which is a common passage for food to the oesophagus and for air to the larynx. At its floor there is the glottis, which opens into the larynx and is guarded by a flap called the epiglottis. The epiglottis prevents food and liquid from entering the windpipe and lungs.

3. Larynx: it is a box present between the pharynx and the trachea, and it is also called the voice box. Two fibrous bands called vocal cords are attached to its inner walls, and the vibrations of these cords produce sounds.

4. Trachea and bronchi: the trachea is a 12 cm long tube attached to the larynx. It ends in two branches called bronchi, and each bronchus enters the lung of its side. The inner walls of the trachea and bronchi are lined with mucus and cilia; mucus traps fine dust and bacteria, and cilia move these particles upward to the pharynx. C-shaped cartilages are present in the wall of the trachea and plate-like cartilages in the walls of the bronchi, which prevent them from collapsing.

5. Bronchioles and alveoli: inside the lungs both bronchi divide again and again into very small tubes called bronchioles. Each bronchiole ends at an alveolar duct which opens into a cluster of thin-walled pouches called alveoli. Each alveolus is surrounded by blood capillaries, and gaseous exchange takes place here.

3. Describe the functions and locations of the bronchus, trachea, and alveoli.

Part	Location	Function
Trachea (windpipe)	A 12 cm long tube in the neck and upper chest, attached above to the larynx and ending below in two bronchi.	It carries air to and from the lungs. Its walls have C-shaped cartilages which keep it open, and its lining of mucus and cilia traps dust and bacteria and moves them upward to the pharynx.
Bronchus (plural: bronchi)	Two branches at the lower end of the trachea; each bronchus enters the lung of its side.	Bronchi carry air into the lungs. They divide again and again into smaller bronchioles, and plate-like cartilages in their walls prevent them from collapsing.
Alveoli (singular: alveolus)	Thin-walled pouches present in clusters at the end of every bronchiole, inside the lungs; each is surrounded by blood capillaries.	They are the site of gaseous exchange: oxygen moves from the air into the blood and carbon dioxide moves from the blood into the air. Their thin, moist walls and very large total surface area make the exchange rapid.

4. What are the possible causes of lung cancer? Explain the harmful effects of smoking apart from lung cancer.

Causes of lung cancer: cancer means an uncontrolled increase in the number of cells in any part of the body, and when masses of non-functional cells are formed in the lungs it is called lung cancer. Its causes are carcinogens such as those present in tobacco smoke, ionizing radiation and viral infections. Passive smoking, that is, inhaling the smoke of another person's cigarette, is also a cause; the smoke from the burning portion of a cigarette is even more dangerous than the smoke exhaled by the smoker. Its symptoms are shortness of breath, coughing (including coughing up blood) and weight loss, and it is treated by surgery, chemotherapy or radiation therapy.

Harmful effects of smoking other than lung cancer:

(i) Smoking causes cancers of the kidneys, oral cavity, larynx, breast, bladder, oesophagus, pancreas and stomach, and benzene present in smoke causes blood cancer (leukaemia).

(ii) Nicotine hardens the walls of the arteries and causes heart problems, and smoking also increases blood cholesterol levels and blood pressure and weakens the blood vessels.

(iii) Carbon monoxide combines with red blood cells and prevents the transport of oxygen, so less oxygen reaches the body tissues.

(iv) Tar is deposited in the lungs and reduces the surface area for gaseous exchange, and other irritants damage the respiratory tract and cause bronchitis, emphysema and asthma.

(v) Smoking stains the teeth, causes bad breath, and increases the risk of tuberculosis two to four times and of pneumonia four times. Non-smokers who inhale another person's smoke have 30 % more chances of heart disease and lung cancer.

D. Inquisitive Questions

1. What can happen to breathing if diaphragm stops working properly?

The diaphragm is a thick muscular sheet below the lungs, and it is one of the two main muscles of breathing. When it contracts and moves down, the volume of the chest increases and air is pulled into the lungs; when it relaxes and moves up, the volume decreases and air is pushed out. If the diaphragm stops working properly, the chest cannot expand and contract normally, so the person has to work very hard to breathe. As a result breathing becomes shallow and difficult, the lungs are not filled properly, less oxygen reaches the blood and carbon dioxide collects in it. In severe cases the person may need a machine (ventilator) to help in breathing.

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2. Why is it dangerous when smoke particles settle inside lungs?

The lungs have a defence system: mucus traps dust and smoke particles, and cilia move them upward to the pharynx so that they can be coughed out. But when a lot of smoke is inhaled again and again, these particles settle in the alveoli, which have no cilia to remove them. There they kill the cells and cause inflammation, and tar reduces the surface area available for gaseous exchange, so the patient feels shortness of breath. The settled particles also contain carcinogens which may start uncontrolled cell division and cause lung cancer, and they make the lungs more liable to infections such as bronchitis and pneumonia.

3. Why does the trachea has C-shaped cartilage rings while bronchioles do not have them?

The trachea is a large, soft tube in the neck and chest, and the pressure of the air inside it changes when we breathe. The C-shaped cartilages in its wall make it strong enough to stay open and prevent it from collapsing when there is no air in it; the C shape also allows the oesophagus behind it to expand when food is swallowed. As the passage divides into bronchi and then into bronchioles, the amount of cartilage decreases and it is completely absent in the smallest bronchioles. The bronchioles are very small, and they are kept open by the pressure of the air inside them. If they had cartilage rings, they could not contract and relax to control the flow of air into the alveoli.

4. During asthma attacks, patients feel harder to exhale than to inhale. Why is it so?

In asthma the bronchi and bronchioles are inflamed, their muscles constrict and extra mucus is produced, so their inner space becomes narrow. During inhalation the intercostal muscles and the diaphragm contract actively and pull the ribcage outward and the diaphragm downward, which helps the air to push the narrowed tubes open. But during exhalation these muscles only relax, so the air has to be pushed out by the elastic recoil of the lungs alone through the narrowed tubes. Moreover, while the person pushes harder to breathe out, the pressure in the chest squeezes the soft bronchioles further. That is why a patient feels it much harder to breathe out than to breathe in, and wheezing (a whistling sound) is heard during exhalation.

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